

2024



KONE Unlocks New

Learn how KONE built a secure, reliable, and escalators using AWS IoT.

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40% reduction

in customer-reported elevator and escalator issues

Greater than 70%

proactive identification of faults

40% fewer

entrapments

99.9% provisioning

success rate

5x increase

in scale of IoT devices

Overview

A global leader in the elevator and escalator industry, [KONE](#) needed robust, highly available Internet of Things (IoT) and analytics services for its growing fleet of smart devices. The company wanted to connect a larger share of its equipment to the cloud, digitize operational processes, and deliver new value-added services to customers. KONE uses advanced IoT technologies from Amazon Web Services (AWS) to improve the flow of people and resources in urban settings for increased comfort, convenience, and security while reducing costs through smart, connected, and energy-efficient offerings.

KONE had built an intelligent predictive maintenance offering, KONE 24/7 Connected Services, on a third-party IoT platform. However, KONE soon sought an alternative solution to accommodate its goal of connecting all 1.6 million pieces of equipment in KONE's maintenance base and using their data to streamline operations and create meaningful urban flow experiences for customers. Using AWS, KONE built a highly available architecture that reduced callouts by up to 40 percent and sparked the development of groundbreaking products.



Opportunity | Using AWS IoT Core to Build a Robust and Scalable Architecture for KONE

Founded in 1910 in Helsinki as a machine repair shop, KONE—pronounced CONE-ay, Finnish for “machine”—sells elevators and escalators in more than 60 countries, with the mission of improving the flow of urban life. The company focuses on servicing and modernizing its own and third-party products, including some elevators that are a century old. It offers IoT-enabled sensors to monitor critical parameters such as usage levels and speed, ingesting 3,000 events per second and sending data to the cloud for algorithms that quickly detect potential problems.

In 2020, KONE decided that—rather than binding KONE 24/7 Connected Services to the IoT platform it had implemented in 2016—it should build an agile, scalable, highly available architecture using a large-scale cloud service provider that could help generate near-real-time insights and provide greater design flexibility. Furthermore, KONE’s IoT technology provider stopped facilitating deployment in China, where KONE continues to have a significant presence.

KONE rebuilt its IoT platform using [AWS IoT Core](#), which helps organizations connect billions of IoT devices and route trillions of messages to AWS services without managing infrastructure. “When we evaluated AWS, we felt like we were entering into a collaboration that went way beyond choosing a technology,” says Ashish Agrawal, chief information officer at KONE. “AWS supported our ambitions, including support commitments, bringing us the best experts, and helping create a go-to-market and co-innovation journey and story.”

Solution | Modernizing KONE’s IoT and Analytics Platform and Reducing Callouts by More than 40% Using AWS Cloud Connectivity

When KONE learned about the benefits of AWS native services, it elected to fully transform and modernize its core technology rather than lift and shift its existing stack. KONE built a secure and stable architecture alongside [AWS Professional Services](#), a global team of experts that helps organizations achieve their desired business outcomes using AWS.

KONE customers can upgrade their legacy equipment with IoT technology–powered hardware and KONE 24/7 Connected Services. Implementing these add-ons, KONE customers remotely activate their equipment and securely connect it to the cloud—a process that is commonly referred to as “provisioning.” To AWS IoT Core, KONE’s provisioning success rate has been close to 100 percent. Using the re



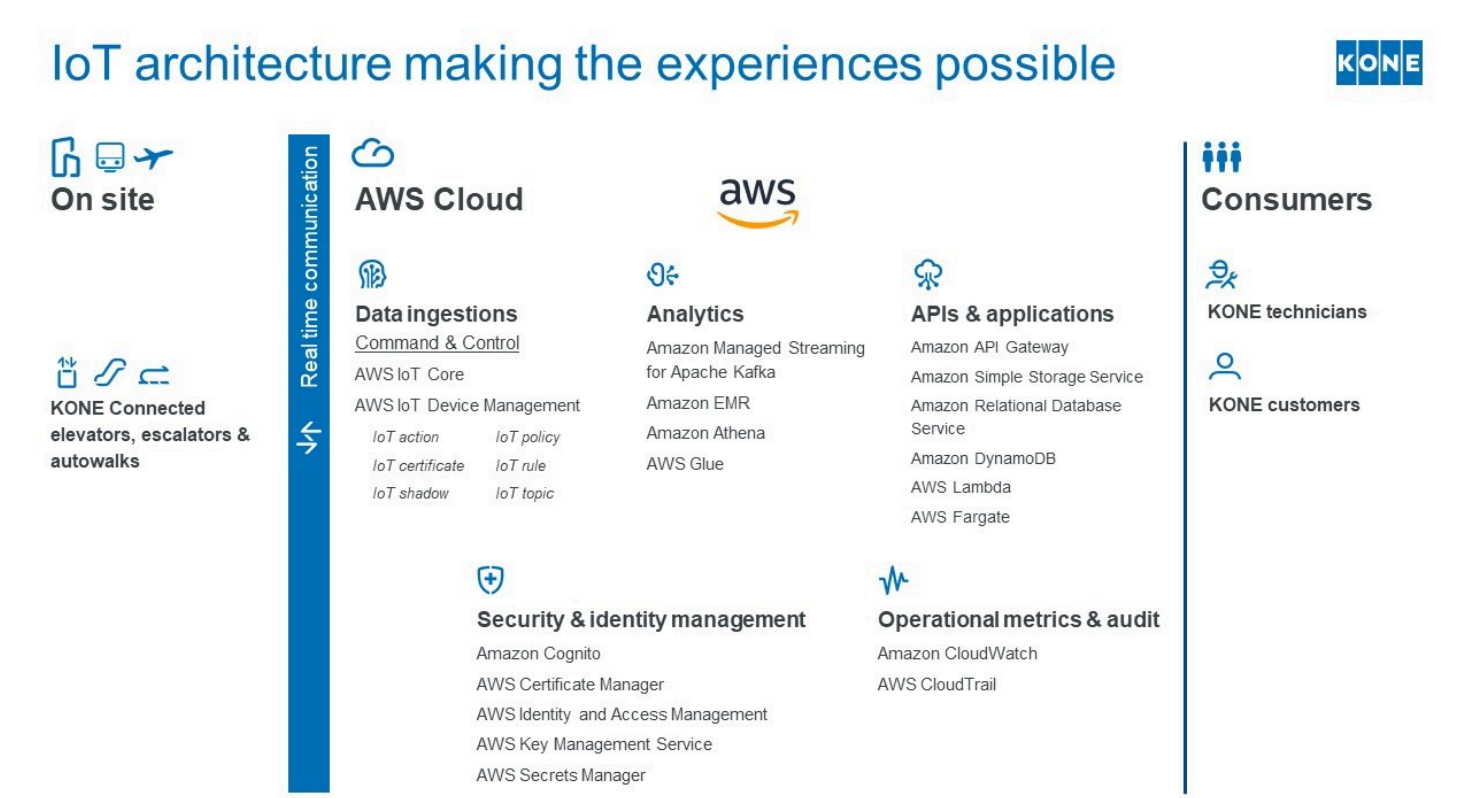
reliable connectivity of AWS IoT Core, KONE also has significantly reduced the time and effort required to resolve outages. “We promise our customers that, around the clock, 24/7, we are monitoring their elevators,” says Agrawal.

Through KONE 24/7 Connected Services, KONE’s customers gain greater visibility into their elevator operations. KONE’s connected systems send information through AWS IoT Core to other AWS services to efficiently monitor and manage their entire fleet of connected offerings. Software updates and equipment reboots are orchestrated through [AWS IoT Device Management](#), which helps organizations register, organize, monitor, and remotely manage IoT devices at scale. Empowered by AWS IoT services, KONE proactively identifies more than 70 percent of faults and has reduced customer-reported elevator and escalator issues by more than 40 percent, lowering costs, improving sustainability metrics, and leading to more efficient operations for itself and for customers. Moreover, underpinning its commitment to the value of robustness, KONE has reduced instances of entrapment—often characterized by the failure of an elevator door with passengers inside—by 40 percent.

KONE has saved significantly on cost as it has scaled to accommodate 500,000 devices—up from 100,000 under its old architecture. “It would have cost us 3 times more and likely would have crashed well before our current scale,” says Agrawal. “We’ve had quality improvements, security improvements, and architectural improvements, and financially we can do so much more on AWS. And the best benefit is the transparency of cost through AWS.”

In addition to its IoT backbone, KONE is building a strong data and analytics foundation on AWS. It stores the data collected from edge devices in [Amazon Simple Storage Service](#) (Amazon S3), an object storage service. Data stored in Amazon S3 serves as a foundation for additional analytics services that KONE can tailor to uncover business insights. Using roughly 70 AWS services, KONE has pivoted toward a DevOps model, in which internal teams manage and provision their own resources.

Architecture Diagram



KONE, which became an [AWS Partner](#) in 2021, has built a robust, reliable, and scalable IoT and analytics platform using AWS IoT services. While it reduces maintenance workflow costs and seamlessly scales its fleet operations, KONE is paving the way for new opportunities to strengthen customer relationships through value-added services. As the next phase of its innovation, KONE is focusing on expanding its use of artificial intelligence and digital twin services to unlock and accelerate value for urban buildings. For example, KONE built a proof of concept of a near-real-time people-flow digital twin for the Helsinki main railway station using [AWS IoT TwinMaker](#), which helps companies optimize operations by creating digital twins of real-world systems.

“The technology core we built on AWS is a backbone on which we can scale, innovate in the market, and differentiate ourselves in the best urban flow experience,” says Agrawal. “It’s not just about delivery of a product. It’s about efficiency, productivity, and transparency. We’re talking about a holistic concept and improved customer and user experience. We want to make daily life simpler.”

About KONE

KONE’s mission is to improve the flow of urban life. Operating in over 60 countries, KONE provides elevators, escalators, and automatic doors, plus solutions for maintenance and modernization to add value to buildings throughout their life cycle.

AWS Services Used

AWS IoT Core

AWS IoT Core lets you connect billions of IoT devices and route trillions of messages to AWS services without managing infrastructure.

[Learn more »](#)

AWS IoT Device Management

AWS IoT Device Management helps you register, organize, monitor, and remotely manage IoT devices at scale.

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AWS IoT TwinMaker



AWS IoT TwinMaker makes it easier for developers to create digital twins of real-world systems such as buildings, factories, industrial equipment, and production lines.

[Learn more »](#)

More Manufacturing Customer Stories

FINLAND



KONE on AWS

KONE's elevator, escalator, and automatic door solutions aim to improve the flow of urban life.

2024

JAPAN



Hitachi on AWS

From leading the way in automotive Internet of Things (IoT) technology to sharpening global supply chain logistics, Hitachi uses Amazon Web Services (AWS) to support innovation and efficiency. It has migrated mission-critical workloads, driven cost savings, and connected millions of vehicles in the cloud.

2021

KOREA



Samsung Electronics Improves Demand...

Using Amazon SageMaker Canvas, Samsung Electronics empowered business analysts without coding experience to forecast demand, freeing

SINGAPORE



KABAM Robotics enhances robot intelligence with...

KABAM Robotics created Smart+ – a cloud-based robot management platform which leverages AWS to enhance the intelligence and concierge



up the data science team to focus on more advanced models.

2023

capabilities of their robots – and REMI – a groundbreaking system equipping their robots with the power of large language models, seamlessly integrated using Amazon Bedrock API, while

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